

The Influence of Perspective on Training Effects in Virtual Reality Public Speaking Training

Fumitaka Ueda*

Yuichiro Fujimoto†

Taishi Sawabe‡

Masayuki Kanbara§

Hirokazu Kato¶

Nara Institute of Science and Technology



Figure 1: Three perspective conditions in the experiment. (a) 1PP: first-person perspective, (b) Back: third-person perspective with a view of the avatar's back, and (c) Front: third-person perspective with a view of the avatar's front.

ABSTRACT

A third-person perspective in virtual reality (VR) based public speaking training enables trainees to observe themselves through self-avatars, potentially enhancing their public speaking skills. This study investigates the influence of perspective on the training effects, specifically changes in public speaking skills through training. In the experiment, VR job interview training was conducted for five days under three perspective conditions. Mock interviews were also performed before and after training, and were assessed by external raters. The results revealed that the training effects were significantly higher in the Front condition concerning verbal communication skills and the overall impression of the interview.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality;

1 INTRODUCTION

Public speaking is making a speech in front of an audience, such as interviews and presentations. In recent years, the use of virtual reality (VR) to enhance public speaking skills has gained attention. In VR, a third-person perspective (3PP) allows users to observe their self-avatars. Prior research shows that using the 3PP in VR public speaking reduces anxiety [1], and facilitates objective self-evaluation when reflecting on one's presentation [5]. These results suggest the usefulness of the 3PP in VR public speaking training. On the other hand, no research has confirmed the impact of perspective on public speaking skills, despite their essential role in audience evaluation. We define changes in public speaking skills before and after training as training effects and investigate the influence of perspective on them. The research question is as follows: **RQ** How does perspective during VR public speaking training influence the training effects?

Thus, we conducted a five-day between-subjects experiment under three perspective conditions using a VR job interview training system. The standard 3PP involves viewing from behind the avatar.

*e-mail: ueda.fumitaka.uz0@is.naist.jp

†e-mail: yfujimoto@is.naist.jp

‡e-mail: t.sawabe@is.naist.jp

§e-mail: kanbara@is.naist.jp

¶e-mail: kato@is.naist.jp

Considering the importance of audience evaluation in public speaking, we included a perspective from the audience side as one of the conditions. To evaluate the training effects, external raters assessed videotaped mock interviews before and after VR training.

2 VR JOB INTERVIEW TRAINING SYSTEM DESIGN

A training system simulating job interviews was made using Unity (see Figure 2). The virtual environment included a user's avatar, generated from a photograph of the user's face using the Headshot Auto in Character Creator 4, and three interviewer agents from Greta [4] and Microsoft Rocketbox. In a VR training, the agent in the middle asked the user questions, provided a brief response to each of the user's answers, and proceeded to the next question.

This system tracked the user's body movements and reflected them on the avatar. The HTC Vive Pro Eye displayed images and tracked the user's gaze and eyelid. The Vive Facial Tracker captured facial expressions. Hand movements were detected by the Leap Motion. Moreover, the user wore four Vive Trackers on each wrist and ankle to control the avatar's body motion using Final IK.

3 EXPERIMENT

3.1 Overview

We examined the influence of perspective on the training effects, specifically changes in public speaking skills for job interviews. There were three conditions: 1PP (first-person perspective), Back (perspective from behind the avatar), and Front (perspective from in front of the avatar) (see Figure 1). A total of 22 graduate students (13 males, 9 females), with a mean age of 23.3 (SD 1.43), participated in the experiment. They were assigned to one of the conditions to ensure an even distribution of confidence and experience with job interviews: 1PP ($n = 7$), Back ($n = 8$), Front ($n = 7$).

3.2 Procedure

Before the experiment, participants were asked to prepare answers to interview questions, covering self-introduction, school day activities, and their research topic. The experiment lasted 5 days, with intervals between dates set to be at least one day apart, averaging 3.8 days (SD 1.4). At the beginning and end of the experiment, participants had face-to-face mock interviews without our VR system and were videotaped for evaluation. The experimenter acted as an interviewer and sat between two mannequins. During the experiment, participants completed one or two 5- to 10-minute VR job interview training sessions each day, totaling eight sessions.

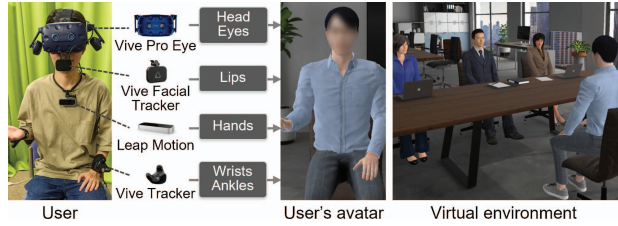


Figure 2: Overview of the VR job interview training system.

3.3 Evaluation of Public Speaking Skills by Others

A scale for assessing public speaking skills for job interviews was used. This scale has three subscales: Prosody, Behavior, and Response. Each category consists of five items. Prosody and Behavior assess non-verbal skills and are based on the Social Performance Rating Scale [3]. Prosody relates to voice (e.g., fluency and intonation), while Behavior concerns observable behaviors (e.g., smile and eye contact). Response focuses on verbal skills and relates to the content of the speech (e.g., logical and concise). Additionally, an Overall item was used to assess the overall impression of the interview. A 7-point Likert scale was employed for all items.

Two raters watched and evaluated the videotaped mock interviews. They did not know the experimental conditions and whether each video was recorded before or after training. The mean of the raters' answers was used as the score for each item. The change in total score for each category before and after training was defined as $Prosody_d$, $Behavior_d$, $Response_d$, and $Overall_d$.

3.4 Hypotheses

In the Back and Front, self-observation may lead to higher evaluations. $Behavior_d$ is expected to be highest in the Front, as participants can observe their avatars in detail. Our hypotheses are:

- H1** $Prosody_d$, $Behavior_d$ and $Response_d$ will be higher in the Back and Front conditions compared to the 1PP condition.
H2 $Behavior_d$ will be highest in the Front condition.

4 RESULTS

Figure 3 displays the training effects for each category by condition. The Shapiro-Wilk test and Bartlett's test were performed to confirm the normality and equality of variances ($\alpha = .05$). Normality was verified in every category except for $Prosody_d$. Equality of variances was observed in every category. Therefore, the Kruskal-Wallis test was conducted for $Prosody_d$, while one-way ANOVAs were employed for $Behavior_d$, $Response_d$, and $Overall_d$. The one-way ANOVAs indicated significant effects of perspective on $Response_d$ ($F(2, 19) = 7.40, p = .004$) and $Overall_d$ ($F(2, 19) = 4.86, p = .020$). None of the other comparisons were significant. Post hoc comparisons using the Tukey HSD test revealed that the mean score of $Response_d$ was significantly higher in the Front ($M = 7.29, SD = 3.24$) than in the 1PP ($M = 2.57, SD = 2.52, p = .008$) and Back ($M = 2.81, SD = 2.00, p = .009$). The mean score of $Overall_d$ was also significantly higher in the Front ($M = 1.57, SD = 0.93$) than in the 1PP ($M = 0.14, SD = 0.75, p = .015$).

5 DISCUSSION

Training effects ($Response_d$ and $Overall_d$) were higher in the Front. Thus, H1 was partially supported regarding the Front. One reason for this difference is the influence of cognitive load. Job interviews require attentiveness to speech content, language usage, and behavior. Additionally, the virtual agents moved independently of the participants' intentions, potentially increasing cognitive load. In the Front, cognitive load was expected to be lower as the agents were not visible from the front. This could have allowed participants to focus on the training. Another factor may be explained by the theory of objective self-awareness [2]. This theory suggests that

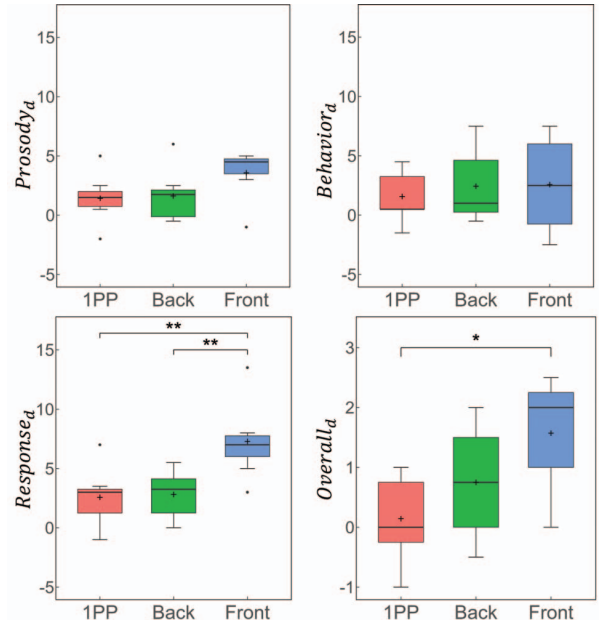


Figure 3: Training effects for each category by condition. * $p < .05$, ** $p < .01$.

when people are focusing on themselves, such as in front of a mirror, they behave according to their own evaluation criteria. Based on this theory, the participants in the Front paid more attention to themselves, which may have led to higher training effects.

On the other hand, the effects of perspective on $Behavior_d$ were not significant. Therefore, H2 was not supported. Taken together with $Prosody_d$, no significant effects of perspective on non-verbal skills were confirmed. In the VR training, the agents only provided simple responses to the participants' answers and did not show facial expressions or gestures. In fact, some participants mentioned that they were not sure if their intentions were conveyed due to the limited reactions from the agents. This inflexibility in communication may have influenced the improvement of non-verbal skills.

6 CONCLUSION

In this study, we investigated the influence of perspective on public speaking skills. The experiment revealed that the training effects were significantly higher in the Front regarding verbal communication skills and the overall impression of the interview. Hence, the use of the avatar's frontal perspective, which is not common in VR, can be beneficial for VR public speaking training. Our findings may contribute to the design of more effective VR training systems.

ACKNOWLEDGMENTS

This work was supported by JST, CREST Grant Number JP-MJCR19A, Japan.

REFERENCES

- [1] A. I. Bellido Rivas, X. Navarro, D. Banakou, R. Oliva, V. Orvalho, and M. Slater. The influence of embodiment as a cartoon character on public speaking anxiety. *Frontiers in Virtual Reality*, 2:695673, 2021.
- [2] S. Duval and R. Wicklund. A theory of objective self awareness. 1972.
- [3] T. Fydrich, D. L. Chambless, K. J. Perry, F. Buergener, and M. B. Beazley. Behavioral assessment of social performance: a rating system for social phobia. *Behaviour Research and Therapy*, 36:995–1010, 1998.
- [4] C. Pelachaud. Greta: Realtime platform to control socio-emotional virtual characters. <https://github.com/isir/greta>.
- [5] H. Zhou, Y. Fujimoto, M. Kanbara, and H. Kato. Virtual reality as a reflection technique for public speaking training. *Applied Sciences*, 11(9):3988, 2021.